

Xijia Tao

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Research Interests

Multimodal deep research and browsing agents; diffusion language models; large language model reasoning and safety.

Education

The University of Hong Kong

Ph.D. in Computer Science, HKU NLP Lab

Hong Kong SAR

Sep 2024 – Jun 2028 (expected)

The University of Hong Kong

B.Eng. in Computer Science

Hong Kong SAR

Sep 2020 – Jun 2024

- Full-tuition scholarship for four years of undergraduate study; Dean’s Honours List, Faculty of Engineering (2021–22).

Research Experience

HKU NLP Lab, The University of Hong Kong

Ph.D. Student

Hong Kong SAR

Sep 2024 – Present

- Develop benchmarks and methods for provenance-aware multimodal browsing agents, including MMSearch-Plus (ICLR 2026).
- Study training and inference for diffusion language models, including adaptive decoding and trajectory-aware Boltzmann modeling.
- Investigate behavioral compression of agent skills for contextual adaptation.

Zhipu AI

AI Research Intern

Jul 2024 – Oct 2024

- Developed methods for synthesizing domain-specific training data and evaluated data quality on reasoning benchmarks by training GLM-4-9B models.

HKU NLP Lab, The University of Hong Kong

Research Assistant

Hong Kong SAR

Sep 2021 – Mar 2024

- Researched vision-language model safety, multilingual transfer, knowledge distillation, and pretrained language models for the legal domain.

Huawei Hong Kong Research Center

Machine Learning Research Intern

Hong Kong SAR

Jun 2023 – Sep 2023

- Implemented multimodal networks in MindSpore for CUDA and Ascend environments and studied language-vision feature alignment for large generative models.

Publications

SoftSkill: Behavioral Compression for Contextual Adaptation

2026

Xijia Tao, Yihua Teng, Xinyu Fu, Ziru Liu, Kecheng Chen, Yuzhi Zhao, Suiyun Zhang, Rui Liu, Lingpeng Kong
arXiv:2606.20333. Paper

Self-Distilled Trajectory-Aware Boltzmann Modeling: Bridging the Training-Inference Discrepancy in Diffusion Language Models

2026

Kecheng Chen, Ziru Liu, Xijia Tao, Hui Liu, Yibing Liu, Xinyu Fu, Shi Wu, Suiyun Zhang, Dandan Tu, Lingpeng Kong, Rui Liu, Haoliang Li
arXiv:2605.11854. Paper

MMSearch-Plus: Benchmarking Provenance-Aware Search for Multimodal Browsing Agents

2026

Xijia Tao, Yihua Teng, Xinxing Su, Xinyu Fu, Jihao Wu, Chaofan Tao, Ziru Liu, Haoli Bai, Rui Liu, Lingpeng Kong
The Fourteenth International Conference on Learning Representations (ICLR 2026). Project | Paper | Code

Beyond Confidence: Adaptive and Coherent Decoding for Diffusion Language Models

2026

Kecheng Chen, Ziru Liu, Xijia Tao, Hui Liu, Xinyu Fu, Suiyun Zhang, Dandan Tu, Lingpeng Kong, Rui Liu, Haoliang Li
The Forty-third International Conference on Machine Learning (ICML 2026). Project | Paper

OmniPlay: Benchmarking Omni-Modal Models on Omni-Modal Game Playing

2025

Fuqing Bie, Shiyu Huang, Xijia Tao, Zhiqin Fang, Leyi Pan, Junzhe Chen, Min Ren, Liuyu Xiang, Zhaofeng He
arXiv:2508.04361. Paper

ImgTrojan: Jailbreaking Vision-Language Models with ONE Image

2025

Xijia Tao, Shuai Zhong, Lei Li, Qi Liu, Lingpeng Kong

Proceedings of NAACL 2025. Paper | Code

Language Versatilists vs. Specialists: An Empirical Revisiting on Multilingual Transfer Ability

2023

Jiacheng Ye, Xijia Tao, Lingpeng Kong

arXiv:2306.06688. Paper